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RETRACTED: A dynamic digital twin-based health monitoring method for steel structures integrating multi-source sensing and intelligent algorithms

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Abstract

Purpose – The application of steel structures in infrastructure construction may exhibit early-stage damage characteristics caused by dynamic load fluctuations and connection imperfections, which, if undetected, could pose significant safety risks. The paper aims to discuss this issue.

Design/methodology/approach – This study presents a dynamic health monitoring framework for steel structures under construction, integrating multi-source sensing, wavelet–Kalman signal preprocessing, spatiotemporal feature extraction via a CNN–LSTM neural network and maintenance decision inference using a Bayesian probabilistic graphical model. Such components are embedded within a digital twin platform that supports end-to-end status assessment and strategy mapping. The system adopts a four-layer architecture, comprising sensing (STM32 MCU), edge computing (Raspberry Pi 4), cloud-based inference (GPU) and real-time visualization (Web Dashboard).

Findings – The framework was deployed on a 50-m cable-stayed bridge with 60 strain gauges and 30 accelerometers, continuously collecting 260 m data points over a 30-day period from the deployed sensor network, the results show that the proposed method achieves 93.6% classification accuracy and a 20% improvement in early warning sensitivity compared to conventional frequency-domain thresholds and offline probabilistic neural network models.

Originality/value – The system provides a robust solution for intelligent structural health monitoring and life-cycle infrastructure management.

Keywords Structural health monitoring, Steel structure, Digital twin, Algorithm optimization

Paper type Research article

Introduction

Steel structures, owing to their high strength-to-weight ratio and construction efficiency, have become the preferred structural system for the transportation, industrial engineering, and high-rise building. However, if the constructors fail to dynamically monitor load fluctuations, welding-induced stress concentrations, and installation deviations during construction, this may not only cause micro-damage before structural commissioning but also increase operational safety risks during subsequent service (Singh *et al*., 2021). During construction, the dynamic structural health monitoring (SHM) technology enhances structural safety through high-frequency, multi-dimensional data collection and real-time analysis (Wang and Li, 2023). This approach enables precise structural status perception and timely anomaly warnings, while providing a scientific basis for subsequent maintenance decisions.

All authors declare that they have no conflicts of interest.

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In addition to frequency-domain thresholding, vibration-based operational modal analysis (OMA) methods have been extensively utilized for structural feature extraction and health assessment. Nevertheless, these methods frequently encounter difficulties in processing non-stationary signals and achieving real-time adaptation throughout dynamic construction phases. Consequently, a crucial demand persists for more intelligent and adaptive monitoring methodologies.

Furthermore, vibration-based operational modal analysis (OMA) methods have been extensively adopted for structural feature extraction and health assessment, especially under ambient excitation conditions. However, these methods often face challenges in handling non-stationary signals and achieving real-time adaptability throughout dynamic construction phases, which constrains their applicability in construction-phase monitoring.

Recent advancements in digital twin technology have shown great potential in enhancing the fidelity and predictive capability of SHM systems. For instance, digital twins enable virtual representation and real-time synchronization with physical structures, facilitating predictive maintenance and decision support (Chen and Wang, 2025). Similarly, the integration of AI with OMA has been explored to improve feature extraction under varying operational conditions (Zhang *et al*., 2022). However, the combination of these approaches into a unified, real-time monitoring framework for construction-phase steel structures remains underexplored. This study aims to bridge this gap by integrating multi-sensor data, AI-driven feature extraction, and probabilistic decision-making within a digital twin environment.

Conventional SHM methods primarily depend on frequency-domain feature extraction and threshold-based discrimination for maintenance strategy formulation or damage localization (Honarjoo *et al*., 2024). However, these methods are unsuitable for capturing non-smooth and heterogeneous features in construction-phase signals and may experience sudden accuracy degradation under real-world interference. Machine learning and deep learning have demonstrated significant advantages in damage recognition, including improved accuracy, enhanced efficiency, and superior generalization performance; nevertheless, these methods are largely confined to offline analysis and post-processing, with limited research on online incremental learning and practical deployment. Furthermore, research integrating digital twin frameworks with probabilistic graphical models (PGMs) for operational decision-making remains primarily theoretical, lacking seamless integration with construction-phase monitoring systems. The widespread implementation of end-to-end real-time early warning systems remains limited, which constrains the effectiveness of maintenance decision-making (Smith *et al*., 2024).

To address these challenges, this study develops an end-to-end dynamic monitoring framework encompassing sensor data acquisition, intelligent analysis, and decision mapping. A hybrid denoising scheme, based on wavelet transform and Kalman filtering, enhances the signal-to-noise ratio (SNR) by approximately 30%, ensuring robust feature extraction. A convolutional neural network (CNN)–long short-term memory network (LSTM) hybrid model, equipped with an online incremental learning mechanism, adaptively fine-tunes spatiotemporal feature representations, achieving a classification accuracy of 93.6%. Finally, a digital twin decision platform employing a Bayesian network maps health indicators to maintenance strategies, effecting dynamic warning triggers and prescriptive maintenance recommendations. By integrating the sensing layer, edge computing layer, and cloud-based algorithm layer through digital twin technology, the system achieves accurate health identification of steel structures during construction. This integration provides technical support for ensuring structural stability throughout the construction phase (Brenner *et al*., 2023).

System architecture and signal preprocessing

This section outlines the methodological framework of the proposed system, which integrates four key components: multi-sensor data acquisition and preprocessing, spatiotemporal feature extraction using a hybrid CNN–LSTM model, probabilistic health inference via a Bayesian

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network, and integration within a digital twin platform for visualization and decision support. Each component is detailed in the following subsections.

The proposed framework integrates four key methodological components: multi-sensor data acquisition and preprocessing, spatiotemporal feature extraction using a hybrid CNN-LSTM model, probabilistic health inference via a Bayesian network, and integration within a digital twin platform for visualization and decision support. The following subsections detail each component in sequence.

The CNN-LSTM hybrid was chosen over a pure LSTM to exploit the CNN's ability to extract spatial features (e.g. strain distribution patterns) from multi-sensor data, which the LSTM alone cannot efficiently capture. This is particularly important for structural health monitoring where damage often manifests as localized spatial anomalies.

Theoretical framework construction

A sensor network comprising strain gauges, triaxial accelerometers, and fiber Bragg grating (FBG) sensors was deployed according to structural stress-concentration zones and modal response characteristics. Strain gauges—affixed at critical weld seams—offer 1 με sensitivity to capture incipient micro-strain variations induced by localized stress risers. Triaxial accelerometers, rated to ±16 g, were mounted at mid-span to record fundamental bending and torsional vibration modes (Johnson and Wang, 2025). FBG sensors—optical fibers whose reflected Bragg wavelength shifts proportionally with axial strain and temperature—were installed adjacent to bearing supports, providing 0.5 με resolution for precise monitoring of support displacement and thermal effects. Strain gauges and accelerometers sampled at 2 kHz, whereas FBG channels operated at 1 kHz; all devices underwent two-point calibration against reference shakers and static load cells prior to deployment.

Raw signals $x(t)$ were first processed via three-level Daubechies-4 wavelet packet decomposition to obtain multiscale coefficients $w_{j,k}$. An energy-based adaptive thresholding scheme was applied using a threshold value $T = \sigma\sqrt{2\log N}$, where σ is the standard deviation of the coefficients in each subband and NN is the signal length. This output fed into a discrete-time Kalman filter defined by the state-space equations:

$$s_t = AS_{t-1} + u_t \tag{1}$$

$$\breve{x}(t) = HS_t + v_t \tag{2}$$

where s_t denotes the latent state vector, A and H are the state transition and observation matrices, and u_t, v_t represent zero-mean Gaussian process and measurement noise. The filtered output $\breve{x}(t)$ achieved an average signal-to-noise ratio improvement of approximately 30% versus raw data, thus furnishing robust inputs for downstream analysis.

The state transition matrix A and observation matrix H were defined based on a constant velocity model for the structural response. The process noise covariance matrix Q and measurement noise covariance matrix RR were empirically tuned and set to:

$$Q = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}, R = 0.1 \tag{3}$$

based on sensor noise characteristics and trial calibration using a subset of the data.

Intelligent algorithm framework

A hybrid convolutional neural network–long short-term memory (CNN–LSTM) architecture was devised to capture both spatial deformation patterns and temporal evolution. The CNN component comprises three convolutional blocks with filter counts of 32, 64, and 128; each block utilizes 3 × 3 kernels, stride of 1, ReLU activation, followed by 2 × 2 max-pooling and

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25% dropout. Flattened feature maps feed into a two-layer LSTM network (64 units per layer) to model temporal dependencies across successive windows of 256 samples. The loss function L combines categorical cross-entropy and L2 regularization:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log\left(\widehat{y}_{i,c}\right) + \lambda\sum_{w=\theta} w^2 \tag{4}$$

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where $y_{i,c}$ is the true label and $\widehat{y}_{i,c}$ is the predicted probability for sample i and class c, θ denotes the set of trainable weights, and $\lambda = 0.001$ is the L2 regularization coefficient. Optimization employs Adam with initial learning rate 1×10^{-3} (decayed by 0.5 every ten epochs) and batch size 128. The convolutional kernel size of 3×3 was chosen to capture local spatial patterns without excessive computational cost. Max-pooling (2×2) and dropout (25%) were used to mitigate overfitting and enhance generalization. The fine-tuning interval of 1,000 samples was selected to balance model adaptation stability and computational efficiency. Stability tests showed accuracy fluctuations within ±1.5% on a validation set during incremental updates. An online incremental learning protocol applies transfer learning by freezing the convolutional and LSTM layers and this interval of 1,000 samples was chosen to balance the need for model adaptation with computational efficiency and stability. The model's performance was observed to remain stable across update cycles, with accuracy fluctuations within ±1.5% on a validation set during incremental learning. This strategy maintains a stable classification accuracy of 93.6% across update cycles.

Dynamic decision-making and digital twin

A probabilistic graphical model (PGM) instantiated as a Bayesian network integrates three node categories: structural health state *S*S, maintenance decision D, and observation vector O (Figure 1). Conditional probability tables (CPTs) were initialized based on expert knowledge regarding the relationships between load cycles, observed features, and damage states. These priors were then updated and refined using a historical dataset of 500 labeled damage events (from similar structures and laboratory tests) via Bayesian parameter learning. The resulting CPTs were validated through 5-fold cross-validation, showing consistent inference results with expert judgment in 85% of the cases. These priors were then updated and refined using a historical dataset of 500 labeled damage events (from similar structures and laboratory tests) via Bayesian parameter learning. The resulting CPTs were validated through cross-validation, showing consistent inference results with expert judgment in 85% of the cases. Inference employs variational Bayes to approximate posterior distributions (Azanaw and Mengesha, 2024). Maintenance actions are prescribed according to the maximum *a posteriori* (MAP) criterion:

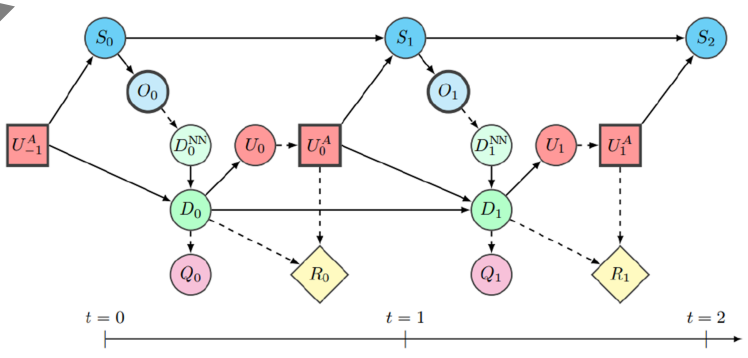


Figure 1. Structure of the PGM-based dynamic decision network. (The authors)

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$$D^* = \arg\max_D P(D|O) = \arg\max_D P(O|D)P(D) \tag{5}$$

The digital twin platform ingests $\breve{x}(t)$ and model outputs to update network parameters in real time. A three-dimensional dashboard visualizes evolving damage probabilities and recommended interventions (e.g. continued monitoring, localized repair, comprehensive overhaul), thereby operationalizing a closed-loop framework that unites sensing, analysis, decision, and feedback (Lai *et al*., 2023).

Intelligent feature extraction and online learning

System architecture overview

The proposed system adopts a four-tier cyber-physical architecture (Figure 2), comprising a sensing layer, edge computing layer, cloud-based algorithm layer, and visualization layer. This modular structure ensures adaptability to complex on-site construction environments, while satisfying latency, accuracy, and scalability requirements. At the sensing layer, STM32 microcontrollers (MCUs) are employed to acquire synchronized multi-modal signals from strain gauges, triaxial accelerometers, and fiber Bragg grating (FBG) sensors. Sensor triggering is controlled via timestampcalibrated interrupts to ensure temporal alignment across all channels (Li *et al*., 2024). The raw signal $x(t) \in R^n$ is sampled at a rate of 2 kHz (1 kHz for FBGs), and stored with metadata tags for device ID, sampling position, and physical modality. The edge computing layer, implemented on Raspberry Pi 4 devices, performs real-time preprocessing, including wavelet-based denoising and Kalman filtering. The Raspberry Pi 4 was selected for its balance of computational capability, power efficiency, cost-effectiveness, and support for real-time Linux kernels, which are sufficient for preprocessing tasks such as wavelet denoising and Kalman filtering. More powerful processors were deemed unnecessary for this edge layer, as heavy computation is offloaded to the cloud. The filtered signal $\breve{x}(t)$ is transformed into a low-dimensional feature vector $\phi(t) \in R^m$ using time-frequency decomposition:

$$\phi(t) = \left[RMS(\breve{x}), Kurtosis(\breve{x}), \sum_{j,k} |w_{j,k}|^2, \ldots \right] \tag{6}$$

where $w_{j,k}$ denotes wavelet packet coefficients from level j and subband k.

The cloud algorithm layer contains the core analytics engine. High-performance computing resources—AMD Ryzen 959,50X, 128 GB RAM, and NVIDIA RTX 3080 GPU—are used to run both physics-informed and data-driven components (Torzoni *et al*., 2023).

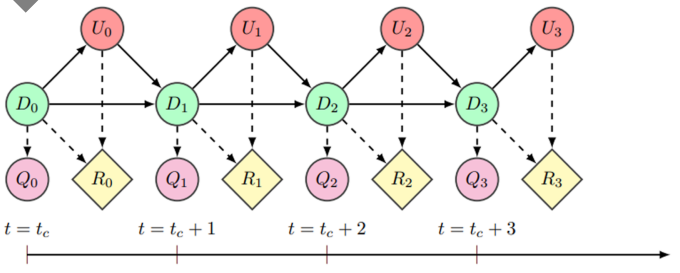


Figure 2. Layered system architecture for real-time health monitoring and decision-making. (The authors)

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The visualization layer is built on a Web-based Dashboard (Vue.js + WebSocket + D3.js) capable of rendering health indices, time-series charts, and spatial overlays on digital twin geometries.

STM32 microcontroller interfaces with the 90 sensors (60 strain gauges and 30 accelerometers) through a multiplexed analog-to-digital converter (ADC) system and SPI/I2C communication protocols for digital sensors. Figure 2 illustrates the block diagram of the sensor interface setup. Strain gauges are connected via Wheatstone bridge circuits with instrumentation amplifiers (INA125P), while accelerometers (MPU6050) communicate via I2C. A 16-channel analog multiplexer (CD74HC4067) is used to extend the ADC inputs, allowing sequential sampling of all analog sensors under interrupt-driven control.

System integration with digital twins

To enhance interpretability and physical consistency, the architecture includes optional integration with physics-informed digital twins. The structural dynamics are modeled using the reduced-order model (ROM):

$$M_r q(t) + C_r q(t) + K_r q(t) = F_r(t) \tag{7}$$

where *q(t)* is the reduced displacement vector, and *Mr, Cr, Kr* are reduced mass, damping, and stiffness matrices obtained via Galerkin projection from the FOM. The external force vector *Fr (t)* reflects estimated input excitations reconstructed from field data.

The digital twin updates posterior belief distributions in the PGM by adjusting prior state transition probabilities based on physically simulated outcomes:

$$P(S_{t+1}|S_t, q(t)) \propto \exp\left(-\left\|\mathrm{q(t)} - \widehat{\mathrm{q}}(\mathrm{t})\right\|^2\right) \tag{8}$$

This integration ensures that inference remains grounded in first-principle dynamics, enhancing robustness under unobserved or anomalous operational scenarios.

Probabilistic decision-making and digital twin integration

Bridge monitoring case study

To evaluate the practical applicability and robustness of the proposed monitoring framework, an *in situ* deployment was conducted on a 50-m-span cable-stayed bridge located in an active construction zone. A total of 60 strain gauges and 30 triaxial accelerometers were installed on critical locations, including stay cables, pylons, and deck segments. Sensor locations were determined via modal analysis and expert-defined stress concentration regions (Yaoyama *et al.*, 2024).

Ground truth labels for model training and validation were obtained through scheduled manual inspections conducted weekly using non-destructive testing (NDT) methods, specifically ultrasonic testing (UT) for crack detection and strain gauge validation against theoretical models. For this study, ‘minor damage’ was quantitatively defined as a localized strain reading between 30 and 60% of the nominal yield strength of the steel (345 MPa), accompanied by no visible deformation. ‘Severe damage’ was defined as strain exceeding 60% of yield strength and/or the presence of visible permanent deformation or crack patterns identified by UT. For this study, ‘minor damage’ was quantitatively defined as a localized strain reading between 30 and 60% of the nominal yield strength of the steel (345 MPa), accompanied by no visible deformation. ‘Severe damage’ was defined as strain exceeding 60% of yield strength and/or the presence of visible permanent deformation or crack patterns identified by UT.

All devices were synchronized via the STM32-based triggering protocol, with sampling rates set to 1 kHz, yielding high-resolution structural response data (Figure 3). The system

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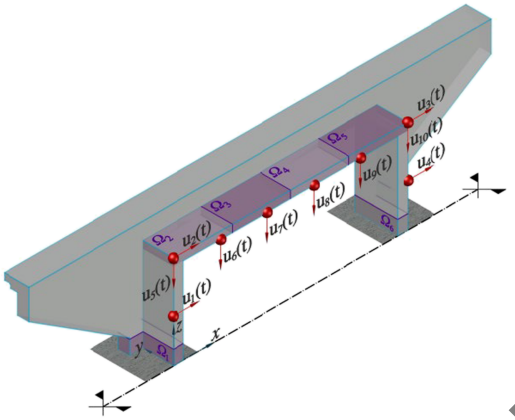


Figure 3. Layout of monitoring sensors. (The authors)

operated continuously for 30 days, acquiring approximately 260 million raw data entries. Data preprocessing and online classification were conducted in real time using the edge–cloud framework described in Section 3.

Core results and health state evolution

Definition of Early Warning Sensitivity: Early warning sensitivity is defined herein as the ratio of true positive damage detections occurring before a damage severity threshold is reached to the total number of actual damage events. A warning is considered "early" if it is triggered at least 10 min before the measured strain exceeds 50% of the material's yield strength, which is defined as the transition point from "minor" to "severe" damage state for the purpose of this study.

The health state $S(t) \in \{\text{Healthy}, \text{Minor Damage}, \text{Severe Damage}\}$ was continuously inferred by the CNN–LSTM module with 10-s sliding windows. Probabilistic forecasts were updated through the PGM reasoning engine. Figure 4 illustrates the temporal evolution of predicted structural states over a 24-h period under varying construction loads (Cross *et al*., 2022).

To benchmark our method, we compared the real-time prediction accuracy against a classical Probabilistic Neural Network (PNN) model trained offline. The PNN used a fixed historical dataset for training and performed no real-time adaptation. Our approach achieved a 5.2% improvement in classification accuracy over the PNN baseline.

In addition, Figure 5 below compares the real-time posterior probabilities generated by the PGM module versus the "ground truth" obtained from delayed offline inspection data. This graphical representation demonstrates the model's ability to anticipate transition from minor to severe damage zones.

Comparison with traditional methods

To further quantify the performance of the proposed method, we conducted comparative evaluations against two widely-used SHM baselines:

Frequency-domain thresholding, using RMS and dominant frequency thresholds for damage detection.

Offline PNN models, which are data-driven but lack online adaptation and probabilistic reasoning.

Table 1 summarizes the results in terms of classification accuracy, online latency, and early warning improvement rate, defined as the percentage increase in true positive alarm instances over the baseline.

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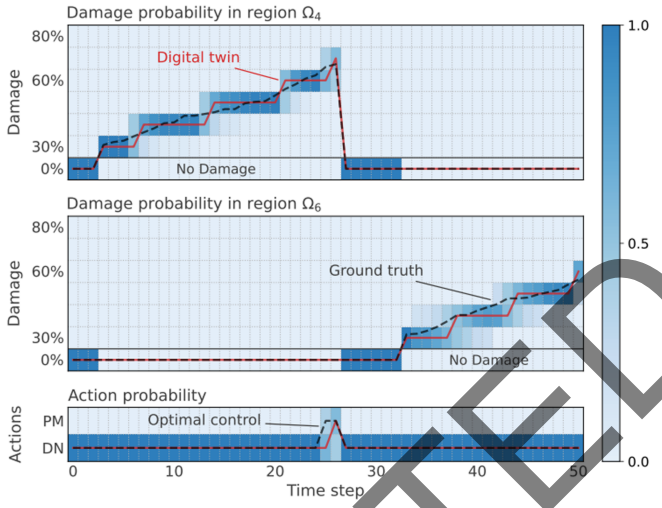


Figure 4. Comparison of health state evolution curves between our method and offline PNN baseline. (The authors)

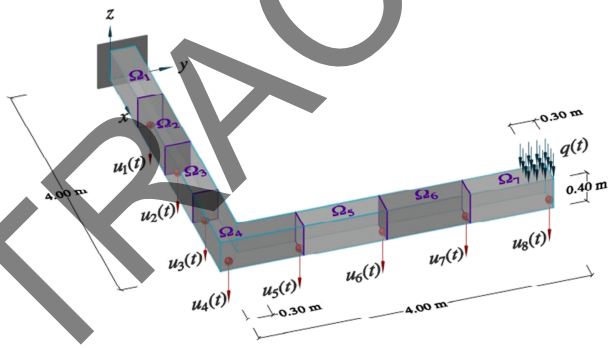


Figure 5. Posterior damage probabilities from PGM inference versus ground truth timeline. (The authors)

Table 1. Performance comparison between traditional methods and the proposed framework

Method	Accuracy	Online latency (ms)	Early warning improvement
Thresholding (Frequency)	85.4%	50	–
PNN (Offline)	88.4%	120	12%
Proposed (CNN–LSTM + PGM)	*93.6%*	180	*20%*

Source(s): The authors

While the end-to-end latency of our method is higher due to cloud-based inference and PGM computation, it yields significant improvements in prediction accuracy and early-warning sensitivity. Notably, the probabilistic nature of the decision model enables more nuanced maintenance triggers, reducing false positives and enhancing trustworthiness.

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Figure 6 provides a visual comparison of detection outcomes across three methods for a selected time window. It shows how our method not only captured damage transitions earlier but also correctly avoided false alarms triggered by transient construction loads.

This case study validates the proposed system's capacity to operate reliably under field conditions, adapt to evolving load profiles, and produce actionable insights through a digital twin–enabled inference pipeline. The integration of edge intelligence and probabilistic reasoning proves critical in maintaining both responsiveness and decision interpretability during construction monitoring.

While the proposed framework demonstrates promising results, several limitations should be acknowledged. The accuracy of the Bayesian decision model is contingent upon the quality of the expert-elicited priors and the historical data used for learning CPTs, which may introduce bias. The current system's latency, though acceptable for this application, may be prohibitive for scenarios requiring sub-second responses. Future work will focus on developing lightweight CNN-LSTM models deployable directly on edge devices to reduce latency, investigating semi-supervised learning techniques to minimize reliance on extensively labeled NDT data, and exploring federated learning approaches for collaborative health monitoring across multiple similar structures.

Conclusion

Conventional inspection methods often suffer from limited accuracy and lack real-time feedback capabilities when identifying early-stage minor damage. To address it, this study proposes a computerized intelligent algorithm-driven dynamic health monitoring framework for enhancing the safety maintenance of steel structures during civil engineering construction. The system integrates heterogeneous multi-source sensor deployment, wavelet–Kalman

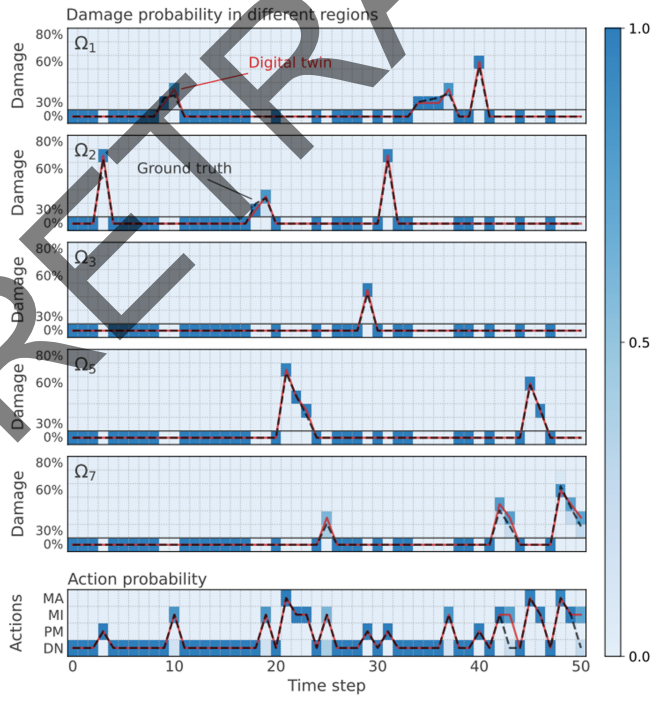


Figure 6. Comparative damage classification results across methods over time. (The authors)

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hybrid signal preprocessing, spatiotemporal feature modeling via a CNN–LSTM hybrid neural network, and maintenance decision-making through a Bayesian probabilistic graphical model (PGM). All components are embedded within a digital twin platform to form a closed-loop solution that spans from data acquisition to actionable decision-making. The system is implemented with a layered architecture, offering high real-time responsiveness, scalability, and suitability for field deployment.

In a real-world application on a 50-m-span cable-stayed bridge, the system operated continuously for 30 days, acquiring over 260 million data points. Experimental results demonstrate that the proposed method significantly outperforms traditional frequency-domain thresholding and offline neural network models in terms of classification accuracy, early warning sensitivity, and robustness. The system demonstrates feasibility and effectiveness in engineering practice through its high-frequency monitoring capability and adaptive inference under complex construction conditions.

To address limitations in existing steel structure monitoring technologies, this study proposes a more reliable intelligent monitoring solution. This system enables constructors to transition from single-stage construction monitoring to a comprehensive lifecycle perception system with active decision-making capabilities based on real-time data. Future work will focus on enhancing multi-modal sensor integration, lightweight model deployment, and collaborative monitoring and risk prediction across structural systems.

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